

The Distribution of Stock Prices

Much of the work that has been done in statistics and related areas regarding the stock market has made the assumption that stock prices are distributed normally, or more specifically, *lognormally*. In actual practice, this is usually an incorrect assumption. For the option strategist, this means that some of the things one might believe about certain option strategies having an advantage over certain *other* option strategies might be incorrect. In this chapter, a number of facts concerning stock price distribution will be brought to light, including how it might affect the option strategist.

MISCONCEPTIONS ABOUT VOLATILITY

Statistics are used to estimate stock price movement (and futures and indices as well) in many areas of financial analysis. Many authors have written extensively about the use of probabilities to aid in choosing viable option strategies. Stock mutual fund managers often use volatility estimates to help them determine how risky their portfolios are. The uses are myriad. Unfortunately, almost all of these applications are wrong! Perhaps *wrong* is too strong a word, but almost all estimates of stock price movement are overly conservative. This can be very dangerous if one is using such estimates for the purposes of, say, writing naked options or engaging in some other such strategy in which volatile stock price movement is undesirable.

As a review for those not familiar with mathematical distributions, the lognormal distribution is what's commonly used to describe stock prices because its shape is

intuitively similar to the way stocks behave—they can't go below zero, they can rise to infinity, and most of the time they don't go much of anywhere. On top of that, the distribution's shape is based on the *historical volatility* of the underlying instrument. In a lognormal distribution (and normal distribution, too), stocks remain within 3 standard deviations of their current price 99.74% of the time. A standard deviation (sigma) is a statistical measure whose absolute distance grows larger with time, and it is something that can be easily calculated for any individual stock or futures contract, using historical prices.

There are great differences between the way stocks *really* behave and the assumptions that many mathematical models make about the way stocks *theoretically* behave. The problem lies in assuming that normal or lognormal distribution predicts stock price movements. Such an assumption does not allow for the occasional wild days that many stocks, some futures, and the relatively rare index undergo. The normal distribution pretty much says that a stock can't rise or fall by more than 3 standard deviations. In fact, according to math, the probability of something that behaves according to the normal distribution (the "classic" bell curve is a normal distribution) moving three standard deviations is 0.0013 (or just a little more than one tenth of one percent). So, if there are 2,500 optionable stocks, say, then one would expect maybe 3 of them to move three standard deviations on any given day.

However, in real market trading, there are routinely moves in stocks of more than 3 standard deviations—some as much as 5 standard deviations or more. Statistically (if the lognormal distribution were correct), one would only expect to see moves of that size maybe once in his lifetime, yet there are five or ten *each day*! Specifically, the normal distribution's probability of something being able to move 8 standard deviations is 0.000000000000000629. This number is so small that one would expect to see only one such occurrence in the known life of the universe, if prices were truly distributed via the normal distribution. If the normal distribution were the correct format for stock prices, such a small probability would indicate that one would not see an 8-standard deviation move until *billions* of trading days had passed. However, one can find several such moves on nearly any trading day, and it is not necessary to use some low-priced, oddball stock that dropped from 1 to .75 or some such nonsense as an example.

If those numbers don't convince you that stocks aren't lognormally distributed, perhaps the following study will. Table 38-1 lists moves that occurred on Monday, April 5, 1999—a day that was somewhat volatile (the Dow was up 174 points).

There were many other substantial moves that day. In the list in Table 38-1, all three that fell in price were on earnings warnings, and the two that rose were just swept up in that day's version of the Internet mania. All in all, 58 stocks had moves of greater than *four* standard deviations on that day! It was not any sort of special day, although it did fall

TABLE 38-1.

Stock	Last Sale	Change	Standard Deviations
Aspect Devt (ASDV)	8	-14.38	-31.2
Axent (ANT)	8	-12	-11.2
Ameritrade (AMTD)	91.63	+29.13	+8.6
CheckPoint (CHKP)	28.75	-10.75	-8.4
Sabre Gp. (TSG)	55	+8.50	+8.0

during earnings warnings season and in the midst of the Internet mania. But the fact that the market is somewhat volatile shouldn't justify all of these huge moves, if one still adheres to the belief that lognormal is the correct distribution.

So, just to make sure that wasn't a bad sample, a low-volatility period was chosen to sample. Since the inception of the CBOE's Volatility Index (VIX) trading, its lowest market volatility readings were in January and July of 1993. The lowest single day was July 25, 1993. On that day, *twelve* stocks had moves of more than four standard deviations. They included some big names, like Adaptec (ADPT), Bethlehem Steel (BS), U.S. Steel (X), Chiquita Brands (CQB), and Novell (NOVL).

The only way to tell how many standard deviations a stock has moved is to use its historical volatility—say, the 20-day historical volatility, for example—in the measurement. Thus, a 4-point move for a nonvolatile stock like Bethlehem Steel in 1993 pales in comparison to Ameritrade's 29-point gain in 1999 (Table 38-1), but both were large moves in terms of *standard deviations*, determined by using each stock's historical volatility.

As another test, prices from October 8, 1998—the day the market bottomed after a severe and rather swift decline brought on by Russian debt problems (it was a very volatile day after several volatile weeks of trading)—were tested to see how many stocks had moves of four standard deviations or more. There were 33, but that seemingly low number reflects the fact that many stocks' 20-day historical volatilities were already well inflated by October 8, 1998. On that day, the Utility Index (UTY) fell over 14 points, which was about 5.5 standard deviations. American Power Conversion (APCC) was *up* over six points that day, to 36.88—a *gain* of over five standard deviations.

Perhaps you might think that these one-day moves overstate things, that over a more prolonged period of time, the lognormal distribution fits better. A study was constructed to measure the volatility over a slightly longer time period. The results of the study not only confirmed our suspicions, but actually were somewhat startling in quantifying just how volatile certain individual issues can be.

The first example comprised the 30-day trading period between October 22, 1999,

and December 7, 1999. There is nothing magical about these dates; that just happened to be the most recent 30-day period for which data was available when the study was conducted.

This particular period started out as a rather normal one in the market—in fact, prices had perhaps been a little less volatile than normal leading into that period. To support that statement, it should be noted that on October 22, the CBOE's Volatility Index (VIX) stood at about 23—a relatively middle-of-the-range level. So it wasn't as if this was an extremely volatile period.

The example is simple enough. The performance of 2,900 optionable stocks was measured to see if, beginning on October 22, 1999, any of them experienced moves of greater than three standard deviations at any time during the 30-day period. The standard deviation was based on the 20-day historical volatility of each individual stock. Obviously, a stock would have to make a much greater move to exceed the 3-standard deviation limit if it waited until the end of the 30-day period to do it, as opposed to making a big move on the first day. So, before reading on, take a guess: How many of the stocks do you think exceeded three standard deviation moves at some point during the 30 days? Remember that the lognormal distribution would predict virtually *no* moves of that size. The answer is in the next paragraph.

There were more stocks that had large upside moves than there were that had large downside moves over the period in question. That isn't too surprising, since the market moved up during that time. The final total showed this: Of the 2,900 stocks, nearly 650 experienced moves of 3 standard deviations or more during the life of the study, including 65 that moved more than six standard deviations. If the lognormal distribution were correct, the two lines in Table 38-2 would be filled with zeroes. This clearly shows stocks during this period didn't conform to the "normal" expectations. The study results are shown in Table 38-2. (Note that " σ " is the greek letter sigma, which mathematicians traditionally use to denote standard deviations, so 3σ means three standard deviations.)

TABLE 38-2.
Stock price movements.

Total Stocks: 2,888			Dates: 10/22/99–12/7/99		
	3σ	4σ	5σ	$>6\sigma$	Total
Upside Moves:	309	116	44	47	516
Downside Moves:	69	29	15	19	132
Total number of stocks moving $\geq 3\sigma$: 648 (22% of the stocks studied)					

The largest move was registered by a stock that jumped from a price of 5 to nearly 12 in about six trading days. One of the bigger downside movers was a stock that fell from about

20 to 8 in a matter of a couple of weeks, with most of the damage occurring in a two-day time period.

Lest you think that this example was biased by the fact that it was taken during a strong run in the NASDAQ market, here's another example, conducted with a different set of data—using stock prices between June 1 and July 18, 1999 (also 30 trading days in length). At that time, there were fewer large moves; about 250 stocks out of 2,500 or so had moves of more than three standard deviations. However, that's still one out of ten—way more than you've been led to expect if you believe in the normal distribution. The results are shown in Table 38-3.

Finally, one more example was conducted, using the least volatile period that we had in our database—July of 1993. Those results are in Table 38-4.

At first glance, it appears that the number of large stock moves diminished dramatically during this less volatile period in the market—until you realize that it still represents 10% of the stocks in the study. There were just a lot fewer stocks with listed options in 1993 than there were in 1999, so the database is smaller (it tracks only stocks with listed options). Once again, this means that there is a far greater chance for large standard deviations moves—about one in ten—than the nearly zero percent chance that the log-normal distribution would indicate.

TABLE 38-3.
More stock price movements.

Total Stocks: 2,447		Dates: 6/1/99–7/18/99			
	3 σ	4 σ	5 σ	>6 σ	Total
Upside Moves:	104	28	13	12	157
Downside Moves:	54	19	7	14	94
Total number of stocks moving $\geq 3\sigma$: 251 (10% of the stocks studied)					

TABLE 38-4.
Stock price movements during a nonvolatile period.

Total Stocks: 588		Dates: 7/1/93–8/17/93			
	3 σ	4 σ	5 σ	>6 σ	Total
Upside Moves:	14	5	1	1	21
Downside Moves:	28	5	3	4	40
Total number of stocks moving $\geq 3\sigma$: 61 (10% of the stocks studied)					

VOLATILITY BUYER'S RULE!

The point of the previous discussion is that stocks move a lot farther than you might expect. Moreover, when they make these moves, it tends to be with rapidity, generally including gap moves. There are not always gap moves, though, over a study of this length. Sometimes, there will be a more gradual transition. Consider the fact that one of the stocks in the study moved 5.8 sigma in the 30 days. There weren't any huge gaps during that time, but anyone who was short calls while the stock made its run surely didn't think it was a gradual advance.

So, what does this information mean to the average option trader? For one, you should certainly think twice about selling stock options in a potentially volatile market (or any market, for that matter, since these large moves are not by any means limited to the volatile market periods). This statement encompasses naked option selling, but also includes many forms of option selling, because of the possibilities of large moves by the underlying stocks.

For example, covered call writing is considered to be "conservative." However, when the stock has the potential to make these big moves, it will either cause one to give up large upside profits or to suffer large downside losses. (Covered call writing has limited profit potential and relatively large downside risk, as does its equivalent strategy, naked put selling.) When these large stock moves occur on the upside, a covered writer is often disappointed that he gave up too much of the upside profit potential. Conversely, if the stock drops quickly, and one is assigned on his naked put, he often no longer has much appetite for acquiring the stock (even though he said he "wouldn't mind" doing so when he sold the puts to begin with).

Even spreading has problems along these lines. For example, a vertical spread limits profits so that one can't participate in these relatively frequent large stock moves when they occur.

What can an option seller do? First, he must carefully analyze his position and allow for much larger stock movements than one would expect under the lognormal distribution. Also, he must be careful to sell options only when they are expensive in terms of implied volatility, so that any decrease in implied will work in his favor. Probably most judicious, though, is that an option seller should really concentrate on indices (or perhaps certain futures contracts), because they are statistically much less volatile than stocks. Hard as it is to believe, futures are less volatile than stocks (although the leverage available in futures can make them a riskier investment overall).

These examples of stock price movement are interesting, but are not rigorously complete enough to be able to substantiate the broad conclusion that stock prices don't behave lognormally. Thus, a more complete study was conducted. The following section presents the results of this research.

THE DISTRIBUTION OF STOCK PRICES

The earlier examples pointed out that, at least in those specific instances, stock price movements don't conform to the lognormal distribution, which is the distribution used in many mathematical models that are intended to describe the behavior of stock and option prices. This isn't new information to mathematicians; papers dating back to the mid-1960s have pointed out that the lognormal distribution is flawed. However, it isn't a terrible description of the way that stock prices behave, so many applications have continued to use the lognormal distribution.

Since 1987, the huge volatility that stocks have exhibited—especially on certain explosive down days such as the Crash of '87 or the mini-crash of April 14, 2000—has alerted more people to the fact that something is probably amiss in their usual assumptions about the way that stocks move. The lognormal distribution “says” that a stock really can't move farther than three standard deviations (whether it's in a day, a week, or a year). Actual stock price movements make a mockery of these assumptions, as stocks routinely move 4, 5, or even 10 standard deviations in a day (not all stocks, mind you, but some—many more than the lognormal distribution would allow for).

In order to further quantify these thoughts, computer programs were written to analyze our database of stock prices, going back over six years. As it turns out, that is a short period of time as far as the stock market is concerned. While it is certainly a long enough time to provide meaningful analysis (there are over 2.5 million individual stock “trading days” in the study), it is a biased period in that the market was rising for most of that time.

THE “BIG” PICTURE

The first part of the analysis shows that the total distribution of stock prices conforms pretty much to what the expectations were for the study, and—not surprisingly—to what others have written about the “real” distribution of stock prices. That is, there is a much greater chance of a large standard deviation move than the lognormal distribution would indicate. The high probabilities on the ends of the distribution are called “fat tails” by most mathematicians and stock market practitioners alike. These “tails” are what get option writers in trouble—and perhaps even leveraged stock owners—because margin buyers and naked writers figure that they will never occur. It is not intuitively obvious to them and to many other stock market participants that stock prices behave in this manner.

The graphs in Figure 38-1 show this total distribution. The top graph is that of the lognormal distribution and the actual distribution, using the data from September 1993 to April 2000—overlaid upon each other. The actual distribution was drawn using 30-day moves (i.e., the number of standard deviations was computed by looking at the stock price on a certain day, and then where it was 30 calendar days later). The x-axis (bottom axis)

shows the number of standard deviations moved. Note that the curves have the shape of a *normal* distribution rather than a *lognormal* distribution, because the x-axis denotes number of standard deviations moved rather than stock prices themselves. For this reason, the term “normal” will be used in the remainder of this section; it should be understood that it is the distribution of standard deviations that is “normal,” while the distribution of the stock prices measured by those standard deviation moves is “lognormal.” The y-axis (left axis) shows the “count”—the number of times out of the 2.5 million data points computed that each point on the x-axis actually occurred (in the case of the “actual” distribution) or could be expected to occur (in the case of the “normal” distribution). The notation on the y-axis shows the actual count divided by 10. So, for example, the highest point (0 standard deviations moved) for the “normal” distribution shows that about 95,000 times out of 2.5 million, you could expect a stock to be unchanged at the end of 30 calendar days.

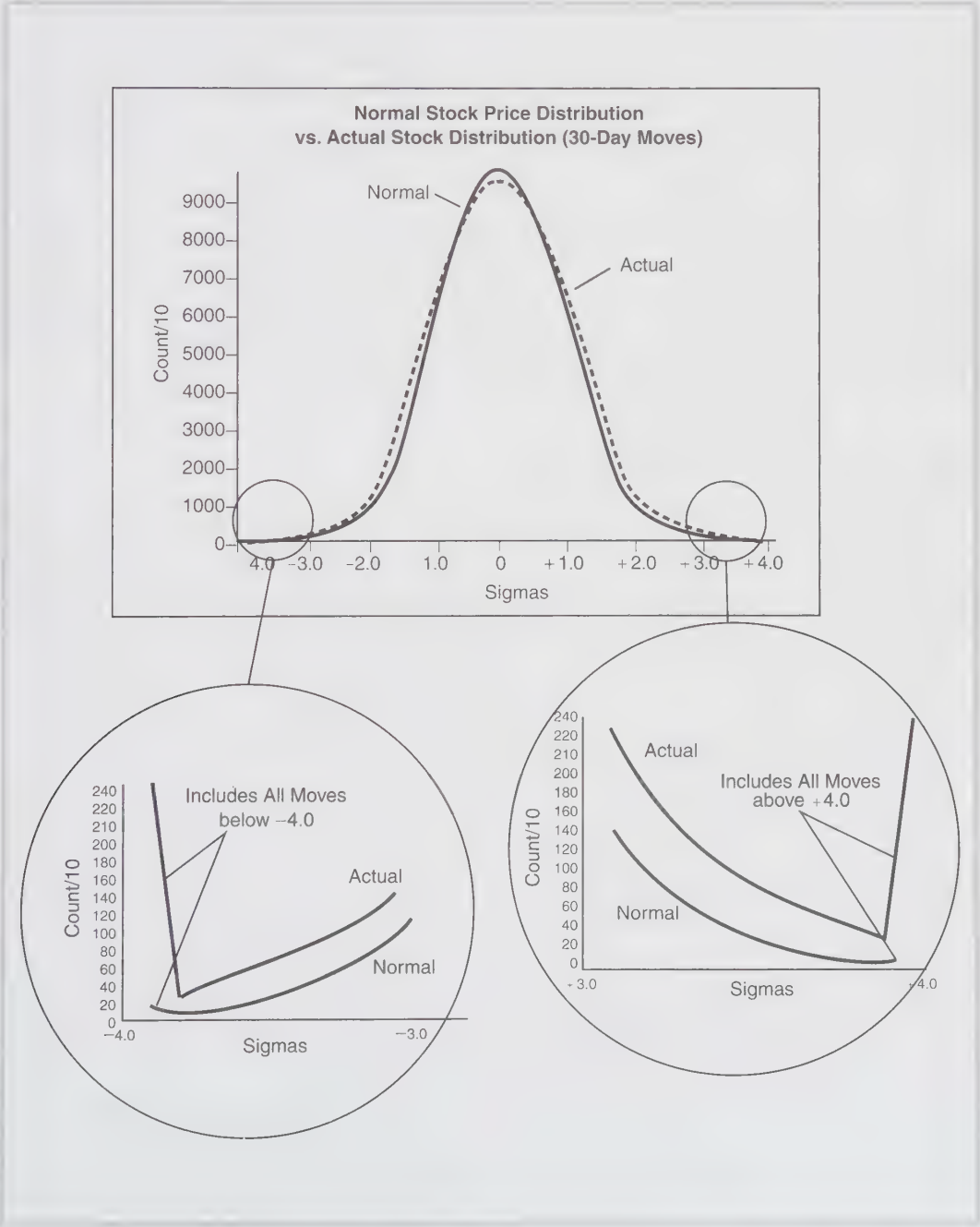
At first glance, it appears that the two curves have almost identical shapes. Upon closer inspection, however, it is clear that they do not, and in fact some rather startling differences are evident.

Fat Tails

Figure 38-1 shows the fat tails quite clearly. Magnified views of the fat tails are provided to show you the stark differences between the theoretical (“normal”) distribution and actual stock price movements. Consider the downside (the lower left circled graph in Figure 38-1). First, note that both the “actual” and “normal” graphs lift up at the end—the leftmost point. This is because the graph was terminated at -4.0 standard deviations, and all moves that were greater than that were accumulated and graphed as the leftmost data point. You can see that the “normal” distribution expects fewer than 200 moves out of 2.5 million to be of -4.0 standard deviations or more (yes, the “normal” distribution *does* allow for moves greater than 3 standard deviations; they just aren’t very probable). On the other hand, actual stock prices—even during the bull market that was occurring during the term of the data in this study—fell more than -4.0 standard deviations nearly 2,500 times out of 2.5 million. Thus, in reality, there was really more than 12 times the chance (2,500 vs. 200) that stocks could suffer a severely dramatic fall, when comparing actual to theoretical distribution. Also notice in that lower left circle that the actual distribution is greater than the normal distribution all along the graph.

The upside fat tail shows much the same thing: Actual stock prices can rise farther than the normal distribution would indicate. At the extreme—moves of $+4.0$ standard deviations or more—there were about 2,000 such moves in actual stock prices, compared with fewer than 100 expected by the normal distribution. Again, a very large discrepancy: twenty-to-one.

FIGURE 38-1.
Stock price distribution is not “normal.”



Inflection Points

If the actual distribution is higher at both ends, it must be lower than the normal distribution *somewhere*, because there are only a total of 2.5 million data points plotted. It turns out in this case that the normal distribution is higher (i.e., is expected to occur more often than it actually does) between -2.5 standard deviations and $+0.5$ standard deviations. Those are the points where the two curves cross over each other—the inflection points. Outside of that range, the actual distribution is more frequent than it was expected to be.

It is probably the case that this data reflected an overly bullish period. That is, actual stock prices rose farther than they were expected to, not necessarily at the tails, but in the intermediate ranges, say between $+0.5$ and $+1.5$ standard deviations. This does not change the results of the study as far as the tails go, but one may not always be able to count on intermediate upside moves being more frequent than predicted.

SIDE BENEFITS OF THIS STUDY

In the course of doing these analyses, a lot of smaller distributions were calculated along the way. One of these is the distribution on any individual trading day that was involved in the study. Now, one must understand that one day's trading yields only about 3,000 data points (there were about 3,000 stocks in the database), so the resulting curve is not going to be as smooth as the ones shown in Figure 38-1. Nevertheless, some days could be interesting. For example, consider the day of the mini-crash, Friday, April 14, 2000. The Dow-Jones Industrials were down 617 that day; the S&P 500 index was down 83 points; and the NASDAQ-100 was down 346. Except for the Crash of 1987, these were the largest single-day declines in history at the time. The distribution graph is shown in Figure 38-2.

First of all, notice how heavily the distribution is skewed to the left; that agrees with one's intuition that the distribution should be on the left when there is such a serious down day as 4/14/2000. Also, notice that the leftmost data point—representing all moves of -4.0 standard deviations and lower, shows that about 750 out of the 2,984 stocks had moves of that size! That is unbelievable, and it really points out just how dangerous naked puts and long stock on margin can be on days like this. No probability calculator is going to give much likelihood to a day like this occurring, but it *did* occur and it benefited those holding long puts greatly, while it seriously hurt others.

In addition to distributions for individual dates, distributions for individual stocks were created for the time period in question. The graph for IBM, using data from the same study as above (September 1993 to April 2000) is shown in Figure 38-3. In the next graph, Figure 38-4, a longer price history of IBM is used to draw the distribution: 1987 to 2000. Both graphs depict 30-day movements in IBM.

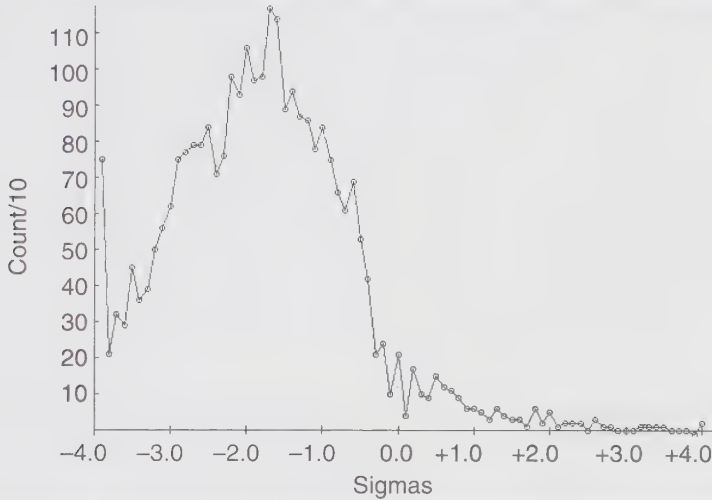
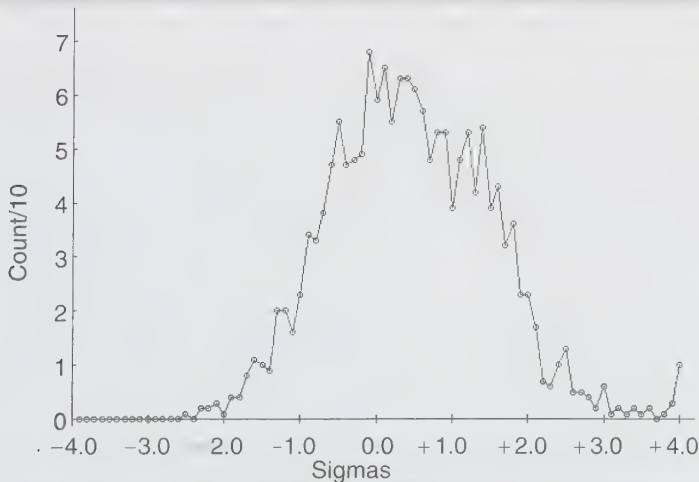
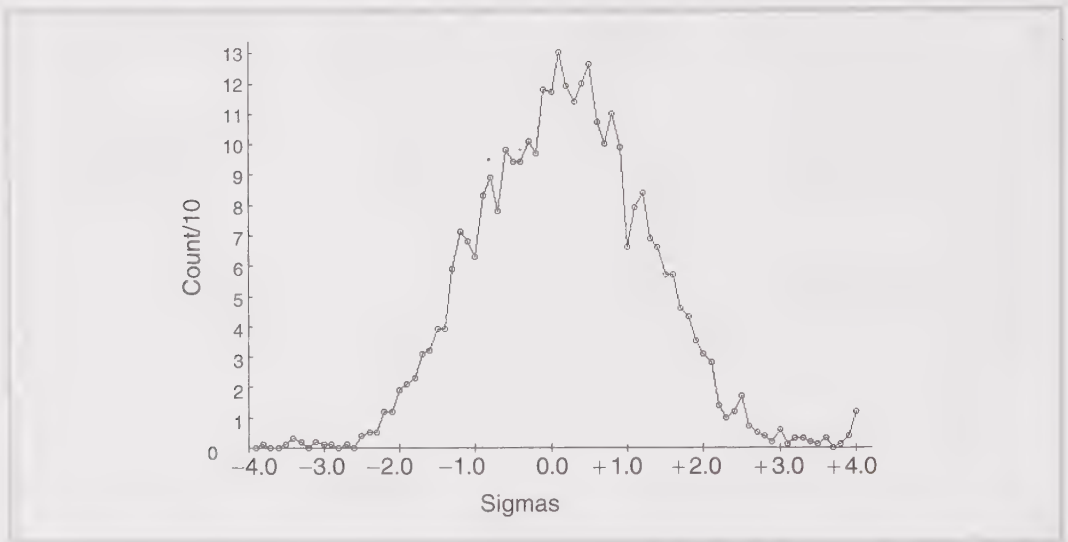
FIGURE 38-2.**Stock price distribution for 4/14/2000 – 2,984 Stocks in Study.****FIGURE 38-3.****Stock price distribution, IBM, 7-year.**

Figure 38-3 perhaps shows even more starkly how the bull market affected things over the six-plus year period, from 1993 to 2000. There are over 1,600 data points for IBM (i.e., daily readings) in Figure 38-3, yet the whole distribution is skewed to the right. It apparently was able to move up quite easily throughout this time period. In fact, the worst

FIGURE 38-4.
Actual stock price distribution, IBM, 13-year.



move that occurred was *one* move of -2.5 standard deviations, while there were about *ten* moves of $+4.0$ standard deviations or more.

For a longer-term look at how IBM behaves, consider the longer-term distribution of IBM prices, going back to March 1987, as shown in Figure 38-4.

From Figure 38-4, it's clear that this longer-term distribution conforms more closely to the normal distribution in that it has a sort of symmetrical look, as opposed to Figure 38-3, which is clearly biased to the right (upside).

These two graphs have implications for the big picture study shown in Figure 38-1. The database used for this study had data for most stocks only going back to 1993 (IBM is one of the exceptions); but if the broad study of all stocks were run using data all the way back to 1987, it is certain that the "actual" price distribution would be more evenly centered, as opposed to its justification to the right (upside). That's because there would be more bearish periods in the longer study (1987, 1989, and 1990 all had some rather nasty periods). Still, this doesn't detract from the basic premise that stocks can move farther than the normal distribution would indicate.

WHAT THIS MEANS FOR OPTION TRADERS

The most obvious thing that an option trader can learn from these distributions and studies is that buying options is probably a lot more feasible than conventional wisdom would

have you believe. The old thinking that selling an option is “best” because it wastes away every day is false. In reality, when you have sold an option, you are exposed to adverse price movements and adverse movements in implied volatility all during the life of the option. The likelihood of those occurring is great, and they generally have more influence on the price of the option in the short run than does time decay.

You might ask, “But doesn’t all the volatility in 1999 and 2000 just distort the figures, making the big moves more likely than they ever were, and possibly ever will be again?” The answer to that is a resounding, “No!” The reason is that the *current* 20-day historical volatility was used on each day of the study in order to determine how many standard deviations each stock moved. So, in 1999 and 2000, that historical volatility was a high number and it therefore means that the stock would have had to move a very long way to move four standard deviations. In 1993, however, when the market was in the doldrums, historical volatility was low, and so a much smaller move was needed to register a 4-standard deviation move. To see a specific example of how this works in actual practice, look carefully at the chart of IBM in Figure 38-4, the one that encompasses the crash of ’87. Don’t you think it’s a little strange that the chart doesn’t show any moves of greater than minus 4.0 standard deviations? The reason is that IBM’s historical volatility had already increased so much in the days preceding the crash day itself, that when IBM fell on the day of the crash, its move was *less* than minus 4.0 standard deviations. (Actually, its one-day move was greater than -4 standard deviations, but the 30-day move—which is what the graphs in Figure 38-3 and 38-4 depict—was not.)

STOCK PRICE DISTRIBUTION SUMMARY

One can say with a great deal of certainty that stocks do *not* conform to the normal distribution. Actually, the normal distribution is a decent approximation of stock price movement *most* of the time, but it’s these “outlying” results that can hurt anyone using it as a basis for a nonvolatility strategy.

Scientists working on chaos theory have been trying to get a better handle on this. An article in *Scientific American* magazine (“A Fractal Walk Down Wall Street,” February 1999 issue) met some criticism from followers of Elliott Wave theory, in that they claim the article’s author is purporting to have “invented” things that R. N. Elliott discovered years ago. I don’t know about that, but I do know that the article addresses these same points in more detail. In the article, the author points out that chaos theory was applied to the prediction of earthquakes. Essentially, it concluded that earthquakes can’t be predicted. Is this therefore a useless analysis? No, says the author. It means that humans should concentrate on building stronger buildings that can withstand the earthquakes, for no one can predict when they may occur. Relating this to the option market,

this means that one should concentrate on building strategies that can withstand the chaotic movements that occasionally occur, since chaotic stock price behavior can't be predicted either.

It is important that option traders, above all people, understand the risks of making too conservative an estimate of stock price movement. These risks are especially great for the writer of an option (and that includes covered writers and spreaders, who may be giving away too much upside by writing a call against long stock or long calls). By quantifying past stock price movements, as has been done in this chapter, my aim is to convince you that "conventional" assumptions are not good enough for your analyses. This doesn't mean that it's okay to buy overpriced options just because stocks can make large moves with a greater frequency than most option models predict; but it certainly means that the buyer of underpriced options stands to benefit in a couple of ways. Conversely, an option seller must certainly concentrate his efforts where options are expensive, and even then should be acutely aware that he may experience larger-than-expected stock price movements while the option position is in place.

So what does this mean for option strategies? On the surface, it means that if one uses the normal (or lognormal) distribution for estimating the probability of a strategy's success, he *may* get a big move in the stock that he didn't originally view as possible. If one were long straddles, that's great. However, if he is short naked options, then there could be a nasty surprise in store. That's one reason why extreme caution should be used regarding selling naked options on stocks; they can make moves of this sort too often. At least with indices, such moves are far less frequent, although the Dow drop of over 550 points in October 1997 was a move of seven standard deviations, and the crash of '87 was about a 16-standard deviation move—which Professor Mark Rubenstein of the University of California at Berkeley says was something that should occur about once in *ten times the life of our current universe!* That's according to lognormal distribution, of course, which we know understates things somewhat, but it's still a big number under any distribution.

There are two approaches that one can take, then, regarding option strategies. One is to invent another method for estimating stock price distributions. Suffice it to say that that is not an easy task, or someone would have made it well-known already. There have been many attempts, including some in which a large history of stock price movements is observed and then a distribution is fitted to them. The problem with accounting for these occasional large price moves is that it is perhaps an even more grievous error to *overestimate* the probabilities of such moves than to underestimate them.

The second approach is to continue to use the normal distribution, because it's fast and accessible in a lot of places. Then, either rely on option buying strategies (straddles, for example) where implied volatility appears to be low—knowing that you have a chance at better results than the statistics might indicate—or adjust your calculations mentally for these large potential movements if you are using option selling strategies.

THE PRICING OF OPTIONS

The extreme movements of the fat tail distribution should be figured into the pricing of an option, but they really are not, at least not by most models. The Black–Scholes model, for example, uses a lognormal distribution. Personally, this author believes that the Black–Scholes model is an excellent tool for analyzing options and option strategies, but one must understand that it may not be affording enough probability to large moves by the underlying.

Does this mean that most options are underpriced, since traders and market-makers are using the Black–Scholes model (or similar models) to price them? Without getting too technical, the answer is that yes, some options—particularly out-of-the-money options—are probably underpriced. However, one must understand that it is still a relatively rare occurrence to experience one of these big moves—it’s just not as rare as the lognormal distribution would indicate. So, an out-of-the-money option might be *slightly* underpriced, but often not enough to make any real difference.

In fact, *futures* options in grains, gold, oil, and other markets that often experience large and sudden rallies display a distinct volatility skew. That is, out-of-the-money *call* options trade at significantly higher implied volatilities than do at-the-money options. Ironically, there is far less chance of one of these hyper-standard-deviation moves occurring in commodities than there is in stocks, at least if history is a guide. So, the fact that some out-of-the-money futures options are expensive is probably an incorrect *overadjustment* for the possibility of large moves.

THE PROBABILITY OF STOCK PRICE MOVEMENT

The distribution information introduced in this chapter can be incorporated into somewhat rigorous methods of determining probabilities. That is, one can attempt to assess the chances of a stock, futures contract, or index moving by a given distance, and those chances can incorporate the fat tails or other *non*-lognormal behavior of prices.

The software that calculates such probabilities is typically named a “probability calculator.” There are many such software programs available in the marketplace. They range from free calculators to completely overpriced ones selling for more than \$1,000. In reality, high-level probability calculation software can be created by someone with a good understanding of statistics, or a program can be purchased for a rather nominal fee—perhaps \$100 or so.

Before getting into these various methods of probability estimation, it should be noted that all of them require the trader to input a volatility estimate. There are only a few other inputs, usually the stock price, target price(s), and length of time of the study.

The volatility one inputs is, of course, an estimate of *future* volatility—something that cannot be predicted with certainty. Nevertheless, any probability calculator requires this input. So, one must understand that the results one obtains from any of these probability calculators is an estimate of what might happen. It should not be relied on as “gospel.”

Additionally, probability calculators make a second assumption: that the volatility one inputs will remain constant over the entire length of the study. We *know* this is incorrect, for volatility can change daily. However, there really isn't a good way of estimating how volatility might change in the course of the study, so we are pretty much forced to live with this incorrect assumption as well.

There is no certain way to mitigate these volatility “problems” as far as the probability calculator is concerned, but one helpful technique is to bias the volatility projection *against* your objectives. That is, be overly conservative in your volatility projections. If things turn out to be better than you estimated, fine. However, at least you won't be overstating things initially. An example may help to demonstrate this technique.

Example: Suppose that a trader is considering buying a straddle on XYZ. The five-month straddle is selling for a price of 8, with the stock currently trading near 40. A probability calculator will help him to determine the chances that XYZ can rise to 48 or fall to 32 (the break-even points) prior to the options' expiration. However, the probability calculator's answer will depend heavily on the volatility estimate that the trader plugs into the probability calculator. Suppose that the following information is known about the historical volatility of XYZ:

10-day historical volatility:	22%
20-day historical volatility:	20%
50-day historical volatility:	28%
100-day historical volatility	33%

Which volatility should the trader use? Should he choose the 100-day historical volatility since this is a five-month straddle, which encompasses just about 100 trading days until expiration? Should he use the 20-day historical volatility, since that is the “generally accepted” measure that most traders refer to? Should he calculate a historical volatility based *exactly* on the number of days until expiration and use that?

To be most conservative, *none* of those answers is right, at least not for the right reasons. Since one is buying options in this strategy, he should use the *lowest* of the above historical volatility measures as his volatility estimate. By doing so, he is taking a conservative approach. If the straddle buy looks good under this conservative assumption, then

he can feel fairly certain that he has not overstated the possibilities of success. If it turns out that volatility is *higher* during the life of the position, that will be an added benefit to this position consisting of long options. So, in this example, he should use the 20-day historical volatility *because it is the lowest of the four choices that he has*.

Similarly, if one is considering the sale of options or is taking a position with a negative vega (one that will be harmed if volatility increases), then he should use the *highest* historical volatility when making his probability projections. By so doing, he is again being conservative. If the strategy in question still looks good, even under an assumption of high volatility, then he can figure that he won't be unpleasantly surprised by a higher volatility during the life of the position.

There have been times when a 100-day lookback period was not sufficient for determining historical volatility. That is, the underlying has been performing in an erratic or unusual manner for over 100 days. In reality, its true nature is not described by its movements over the past 100 days. Some might say that 100 days is not enough time to determine the historical volatility in any case, although most of the time the four volatility measures shown above will be a sufficient guide for volatility.

When a longer lookback period is required, there is another method that can be used: Go back in a historical database of prices for the underlying and compute the 20-day, 50-day, and 100-day historical volatilities for *all* the time periods in the database, or at least during a fairly large segment of the past prices. Then use the *median* of those calculations for your volatility estimates.

Example: XYZ has been behaving erratically for several months, due to overall market volatility being high as well as to a series of chaotic news events that have been affecting XYZ. A trader wants to trade XYZ's options, but needs a good estimate of the "true" volatility potential of XYZ, for he thinks that the news events are out of the way now. At the current time, the historical volatility readings are:

20-day historical:	130%
50-day historical	100%
100-day historical	80%

However, when the trader looks farther back in XYZ's trading history, he sees that it is not normally this volatile. Since he suspects that XYZ's recent trading history is not typical of its true long-term performance, what volatility should he use in either an option model or a probability calculator?

Rather than just using the maximum or minimum of the above three numbers

(depending on whether one is buying or selling options), the trader decides to look back over the last 1,000 trading days for XYZ. A 100-day historical volatility can be computed, using 100 consecutive trading days of data, for 901 of those days (beginning with the 100th day and continuing through the 1,000th day, which is presumably the current trading day). Admittedly, these are not completely unique time periods; there would only be ten non-overlapping (independent) consecutive 100-day periods in 1,000 days of data. However, let's assume that the 901 periods are used. One can then arrive at a *distribution* of 100-day historical volatilities. Suppose it looks something like this:

Percentile	100-Day Historical
0 th	34%
10 th	37%
20 th	43%
30 th	45%
40 th	46%
50 th	48%
60 th	51%
70 th	58%
80 th	67%
90 th	75%
100 th	81%

In other words, the 901 historical volatilities (100 days in each) are sorted and then the percentiles are determined. The above table is just a snapshot of where the percentiles lie. The range of those 901 volatilities is from 34% on the low side to 81% on the high side. Notice also that there is a very flat grouping from about the 20th percentile to the 60th percentile: The 100-day historical volatility was between 43% and 51% over that entire range. The *median* of the above figures is 48%—the 100-day volatility at the 50th percentile.

Referring to the early part of this example, the current 100-day historical is 80%, a very high reading in comparison to what the measures were over the past 1,000 days, and certainly much higher than the median of 48%.

One could perform similar analyses on the 1,000 days of historical data to determine where the 10-day, 20-day, and 50-day historical volatilities were over that time. Those, too, could be sorted and arranged in percentile format, using the 50th percentile (median)

as a good estimate of volatility. After such computations, the trader might then have this information:

Using 1,000 days of data:

Median 100-day historical volatility: 48%
Median 50-day historical volatility: 49%
Median 20-day historical volatility: 52%
Median 10-day historical volatility: 49%

If these were all the data that one had, then he would probably use a volatility estimate of 48% or so in his option models or probability calculators. Of course, this is starkly different from the current levels of historical volatility (shown at the beginning of this example). So, one must be careful in assessing whether he expects the stock to perform more in line with its longer-term (1,000 trading days) characteristics or if there is some reason to think that the stock's behavior patterns have changed and the higher, more recent volatilities should be used.

The pertinent volatilities to consider, then, in a strategy analysis are the medians as well as the current figures. If the trader were going to be buying options in his strategy, should he use the minimum of the volatilities shown, 48%? Probably. However, if he's a seller of options, should he use the maximum, 130%? That might be a little *too* much of a penalty, but at least he would feel safe that if his volatility selling position had a positive expected return with that high a volatility projection, then it must truly be an attractive position.

In an analysis like that shown in this example, there is nothing magical about using 1,000 trading days. Perhaps something like 600 trading days would be better. The idea is to use enough trading days to bring in some historic data to counterbalance the recent, erratic behavior of the stock.

Among other things, this example also shows that volatilities are unstable, no matter how much work and mathematics one puts into calculating them. Therefore, they are at best a fragile estimate of what might happen in the future. Still, it's the best guess that one can make. The trader should realize, though, that when volatilities are this disparate when comparing recent and more distant activity, the results of any mathematical projections using those volatilities should not be relied upon too heavily. Those results will be just as tenuous as the volatility projections themselves.

Of course, in any case, the actual volatility that occurs while the position is in place may be even more unfavorable than the one the trader used in his initial analysis. There

is nothing that one can do about that. But if you choose what appears to be a somewhat unfavorable volatility, and the position still looks good under those assumptions, then it is likely that the trader will be pleasantly surprised more often than not—that actual volatility during the life of the position will tend to be more in his favor than not.

In a recent chapter, the various methods of trying to predict volatility were outlined, using either historical volatility, implied volatility, a moving average of either of those, or even GARCH volatility. None of these will predict with certainty what is going to happen in the future. Hence, the prediction of volatility is necessarily vague at best.

In addition to the vagaries of estimating volatility, the probability calculators will return an answer that represents the probability of something happening “in the long run.” That is, if the same scenario were to arise many, many times, the answer is relevant to how many times the stock would move to the indicated target price. This is small solace if one happens to be caught in the vortex of the Crash of '87, for example. So, just remember that these probability calculators are tools that can help you in assessing the relative risks of similar positions (evaluating various naked option sales, say), but the resulting stock movement in any one case can be quite different from what any probability calculator describes as the chances of that move actually happening.

THE ENDPOINT CALCULATION

The following paragraphs describe how the various probability calculation mechanisms work. The simplest and most straightforward probability calculation has already been presented in Chapter 28 on mathematical applications. It was included in the section on “expected returns” in that chapter. The formula is presented here again, for completeness.

The formula gives the probability of a stock, which is currently at price p , being below some other price, q , at the end of the time period. The lognormal distribution is assumed.

Probability of stock being below price q at end of time period, t

$$P(\text{below}) = N\left(\frac{\ln\left(\frac{q}{p}\right)}{v_t}\right)$$

where

N = cumulative normal distribution

p = current price of the stock

q = price in question

$\ln$ = natural logarithm for the time period in question

If one is interested in computing the probability of the stock being above the given price, the formula is

$$P(\text{above}) = 1 - P(\text{below})$$

In the above formula, $v_t = v\sqrt{t}$ where t is time to expiration in years and v is annual volatility, as usual.

This formula is quite elementary for predictive purposes, and it is used by many traders. This calculator can be found for free at the website www.optionstrategist.com. Its main problem is that it gives the probability of the stock being above or below the target price *at the end of the time period, t* . That's not a totally realistic way of approaching probability analysis. Most option traders are very concerned with what happens to their positions *during* the life of the option, not just at expiration.

Example: Suppose a trader is a seller of naked put options. He sells OEX October 550 puts naked, with OEX currently trading at 600. He would not normally just walk away from this position until October expiration, because of the large risk involved with the sale of a naked option. There are essentially three scenarios that can occur:

1. OEX might *never* fall to 550 by expiration. In this case, he would have a very comfortable trade that was never in jeopardy, and the options would expire worthless.
2. OEX might fall below 550 and remain there until expiration. In this case, he would surely have a loss unless OEX were just a tiny bit below 550.
3. OEX might fall below 550 at some time between when the trade was established and when expiration occurred, but then subsequently rally back above 550 by the time expiration arrived.

An experienced option trader would almost certainly adjust if scenario 3 arose, in order to prevent large losses from occurring. He might roll his naked puts down and out to another strike, or he might just close them out altogether. However, it is unlikely that he would do *nothing*.

The simple probability calculator formula shown above does *not* take into account the trader's third scenario. Since it is only concerned with where the stock is at expiration of the options, only scenarios 1 and 2 apply to it. Hence the usage of this simple calculator is not really descriptive of what might happen to a trade *during* its lifetime.

Let's assign some numbers to the above trade, so that you might see the difference. Suppose that the volatility estimate is 25%, there are 30 days until expiration, and

the prices are as stated in the previous example: OEX is at 600, and the striking price of the naked put being sold is 550. The resulting probabilities might be something like this:

Scenario	Actual Probability of Occurrence
1. OEX never falls below 550	67%
2. OEX falls below 550 and remains there	19%
3. OEX falls below 550 but rallies later	14%

The probabilities stated above are the “real” probabilities of the three various scenarios occurring. However, if one were using the simple probability calculator presented above, he would only have the following information:

Probability of OEX being above 550 at expiration: 81%
Probability of OEX being below 550 at expiration: 19%

So, with the simple calculator, it looks like there’s an 81% chance of a worry-free trade. Just sit back and relax and let the option expire worthless. However, in real life—as shown by the previous set of probabilities, there’s only a 67% chance of a worry-free trade. The difference—the other 14%—is the probability of the third scenario occurring (OEX falls below 550, but rallies back above it by expiration). The simple probability calculator doesn’t account for that scenario at all.

Hence, most serious traders don’t use the simple model. Does that mean it’s not useful at all? No, it is certainly viable as a comparative tool; for example, to compare the chances of the OEX put expiring worthless versus those of another put sale being considered, perhaps something in a stock option. However, better analyses can be undertaken.

Before leaving the scenario of the simple probability calculator, one more point should be made. It has been mentioned earlier in this book that the delta of an option is actually a fairly good estimate of the probability of the option being in-the-money at its expiration date. Thus, the delta and the simple endpoint probability calculator shown above attempt to convey the same information to a trader. In reality, because of the fact that implied volatility might be different for various strikes (a volatility skew), especially in index options, the delta of the option might not agree exactly with the probability calculator. Even so, the delta is a quick and dirty way of estimating the probability of the stock being above the strike price (in the case of call options) or below the strike price (in the case of put options) at expiration.

THE “EVER” CALCULATOR

Having seen the frailties of the endpoint calculator, the next step is to try to design a calculator that can estimate the probability of the stock *ever* hitting the target price(s) *at any time* during the life of the probability study, usually the life of an option. It turns out that there are a couple of ways to approach this problem. One is with a Monte Carlo analysis, whereby one lets a computer run a large number of randomly generated scenarios (say, 100,000 or so) and counts the number of times the target price is hit. A Monte Carlo analysis is a completely valid way of estimating the probability of an event, but it is a somewhat complicated approach.

In reality, there is a way to create a single formula that can estimate the “ever” probability, although it is not any easy task either. In the following discussion, I am borrowing liberally from correspondence with Dr. Stewart Mayhew, Professor of Mathematics at the University of Georgia. For proprietary reasons, the exact formula is not given here, but the following description should be sufficient for a mathematics or statistics major to encode it. If one is not interested in implementing the actual formula, the calculation can be obtained through programs sold by McMillan Analysis Corp. at www.optionstrategist.com.

This discussion is quite technical, so readers not interested in the description of the mathematics can skip the next paragraph and instead move ahead to the next section on Monte Carlo studies.

These are the steps necessary in determining the formula for the “ever” probability of a stock hitting an upside target at any time during its life. First, make the assumption that stock prices behave randomly, and perform at the risk-free rate, r . Mathematicians call random behavior “Brownian motion.” There are a number of formulae available in statistics books regarding Brownian motion. If one is to estimate the probability of reaching a maximum (upside target) point, what is needed is the known formula for the *cumulative density function (CDF)* for a running maximum of a Brownian motion. In that formula, it is necessary to use the lognormal function to describe the upside target. Thus, instead of using the actual target price in the CDF formula, one substitutes $\ln(q/p)$, where q is the target price and p is the current stock price.

The “ever” probability calculator provides much more useful information to a trader of options. Not only does a naked option seller have a much more realistic estimate of the probability that he’s going to have to make an adjustment during the life of an option, but the option buyer can find the information useful as well. For example, if one is buying an option at a price of 10, say, then he could use the “ever” probability calculator to estimate the chances of the stock trading 10 points above the striking price at any time during the life of the option. That is, what are the chances that the option is going to at least break even? The option buyer can, of course, determine other things too, such as the probability

that the option doubles in price (or reaches some other return on investment, such as he might deem appropriate for his analysis).

THE MONTE CARLO PROBABILITY CALCULATOR

Up to this point, the calculators we have discussed are subject to the limitations described earlier—mainly, that they rely heavily on one's volatility estimate, that they assume the volatility will remain constant over time, and that they assume a lognormal distribution. The early part of this chapter was spent explaining that the lognormal distribution is *not* the real distribution that stock prices adhere to. So, what we'd like to see in a probability calculator is one that could adjust for various volatility scenarios as time passed and one in which the assumed distribution of stock prices was *not* lognormal.

When one starts to make these sorts of assumptions, I do not believe there is a single formula that can be derived for the probability calculations. Rather, what is known as a Monte Carlo simulation must be undertaken. Essentially, one “tells” the computer what he is trying to simulate. It could be any number of things in real life, perhaps the rocket engine components in a NASA space shuttle, or the operation of an internal combustion engine, or the movement of a stock. As long as the process can be described, it can be simulated by a computer. Then, the computer can run a large number of those simulations to determine the answers to such things as “What is the failure rate of the NASA engine components?” or “How long can the internal combustion engine go without an oil change?” or “What is the probability of the stock trading at a certain target price?” The Monte Carlo simulation technique can be thought of as letting the computer run through the simulation a *lot* of times and counting how many times a certain outcome occurs. If the number of trials (simulations) is large enough and the model is good enough, then the resulting count divided by the number of trials undertaken is a good probability estimate of the said event occurring. The reason one runs a lot of trials is that over a large number of trials, the frequency with which an event occurs will approximate the actual probability of its occurrence for a single trial—the single trial being your trade, for example.

The next three paragraphs describe the general process necessary for constructing a stock probability calculator using a Monte Carlo simulation. Again, this is fairly technical, so if the reader is not interested in the background behind the mathematics, then skip ahead three paragraphs. In the case of a stock probability calculator, the Monte Carlo simulation can be undertaken as follows.

We know what the distribution of stock prices looks like. The fat tails can be built into the distribution if one wants to simulate real life. See Figure 38-1 for both the lognormal distribution and the actual distribution. It's a simple matter to tell the computer this information. For example, recall that 2.5 million points went into making up Figure 38-1. In the actual distribution in Figure 38-1, about 92,000 (or 3.68%) of them resulted

in the stock being unchanged. Also, only about 2,500 of them, or 1/10th of one percent, resulted in a move of -4.0 standard deviations or more. Those percentages, along with all of the others, would be built into the computer, so that the total distribution accounts for 100% of all possible stock movements.

Then, we tell the computer to allow a stock to move randomly in accordance with whatever volatility the user has input. So, there would be a fairly large probability that it wouldn't move very far on a given day, and a very small probability that it would move three or more standard deviations. Of course, with the fat tail distribution, there would be a larger probability of a movement of three or more standard deviations than there would be with the regular lognormal distribution. The Monte Carlo simulation progresses through the given number of trading days, moving the stock cumulatively as time passes. If the stock hits the break-even price, that particular simulation can be terminated and the next one begun. At the end of all the trials (100,000 perhaps), the number in which the upside target was touched is divided by the total number of trials to give the probability estimate.

Is it really worth all this extra trouble to evaluate these more complicated probability distributions? It seems so. Consider the following example:

Example: Suppose that a trader is considering selling naked puts on XYZ stock, which is currently trading at a price of 80. He wants to sell the November 60 puts, which expire in two months. Although XYZ is a fairly volatile stock, he feels that he wouldn't mind owning it if it were put to him. However, he would like to see the puts expire worthless. Suppose the following information is available to him via the various probability calculators:

Simple "end point" probability of $XYZ < 60$ at expiration:	10%
Probability that XYZ ever trades < 60 (using the lognormal distribution):	20%
Probability that XYZ ever trades < 60 (using the fat tail distribution):	22%

If the chances of the put never needing attention were truly only 10%, this trader would probably sell the puts naked and feel quite comfortable that he had a trade that he wouldn't have to worry too much about later on. However, if the *true* probability that the put will need attention is 22%, then he might *not* take the trade. Many naked option sellers try to sell options that have only probabilities of 15% or less of potentially becoming troublesome.

Hence, the choice of which probability calculation he uses can make a difference in whether or not a trade is established.

Other strategies lend themselves quite well to probability analysis as well. Credit spreaders—sellers of out-of-the-money put spreads—usually attempt to quantify the

probability of having to take defensive action. Any action to adjust or remove a deeply out-of-the-money put credit spread usually destroys most or all of its profitability, so an accurate initial assessment of the probabilities of having to make such an adjustment is important.

Option buyers, too, would benefit from the use of a more accurate probability estimate. This is especially true for neutral strategies, such as straddle or strangle buying, when the trader is interested in the chances of the stock being able to move far enough to hit one or the other of the straddle's break-even points at *some* time during the life of the straddle.

The Monte Carlo probability calculation can be expanded to include other sorts of distributions. In the world of statistics, there are many distributions that define random patterns. The lognormal distribution is but one of them (although it is the one that most closely follows stock prices movements in general). Also, there is a school of thought that says that each stock's individual price distribution patterns should be analyzed when looking at strategies on that stock, as opposed to using a general stock price distribution accumulated over the entire market. There is much debate about that, because an individual stock's trading pattern can change abruptly; just consider any of the Internet stocks in the late 1990s and early 2000s. Thus, a probability estimate based on a single stock's behavior, even if that behavior extends back several years, might be too unreliable a statistic upon which to base a probability estimate.

In summary, then, one *should* use a probability calculator before taking an option position, even an outright option buy. Perhaps straight stock traders should use a probability calculator as well. In doing so, though, one should be aware of the limitations of the estimate: It is heavily biased by the volatility estimate that is input and by the assumption of what distribution the underlying instrument will adhere to during the life of the position. While neither of those limitations can be overcome completely, one can mitigate the problems by using a conservative volatility estimate. Also, he can look at the results of the probability calculation under several distributions (perhaps lognormal, fat tail, and the distribution using only the past price behavior of the underlying instrument in question) and see how they differ. In that case, he would at least have a feeling for what *could* happen during the life of the option position.

EXPECTED RETURN

The concept of expected return was described in the chapter on mathematical applications. In short, expected return is a position's expected profit divided by its investment (or *expected* investment if the investment varies with stock price, as in a naked option position or a futures position). The crucial component, though, is expected profit.

Expected profit is computed by calculating the profitability of a position at a certain stock price times the *probability* of the stock *being* at that price, and summing that multiple over all possible stock prices. When the concept was first introduced, the “probability of the stock being at the price” was given as what we now know is the “endpoint” probability. In reality, a much better measure of the expected profit of a position can be obtained by using one of the more advanced probability estimation models presented above.

In generalized expected return studies done using the fat tails Monte Carlo simulation, certain general conclusions can be drawn about some strategies.

- A bull spread is an inferior strategy when the options are fairly priced, no matter which distribution is assumed. This more or less agrees with observations that have been made previously regarding the disappointments that traders often encounter when using vertical spreads.
- While covered writing might seem superior to stock ownership under the lognormal distribution, the two are about equal under a fat tail distribution.
- Most startling, though, is the fact that option buying strategies fare much, much better under a fat tail distribution than a lognormal one. This most clearly demonstrates the “power” of the fat tail distribution: A limited-risk investment with unlimited profit potential can be expected to perform very well if the fat tails are allowed for.

Using the lognormal distribution more or less represents the conventional wisdom regarding option strategies—the one that many brokers promote: “Don’t buy options, don’t mess with spreads, either buy stocks or do covered call writes.” The fat tail distribution column stands much of that advice on its head. In real life (as demonstrated by the fat tail distribution), strategies with limited profit potential and unlimited or large risk potential are inferior strategies.

One should be aware that the phrase “expected return” is used in many quasisophisticated option analyses (and even in analyses not using options). Many investors accept these “returns” on blind faith, figuring that if they’re generated by a computer, they must be correct. In reality, they may not be representative, even for comparisons.

SUMMARY

This chapter has demonstrated that probability analysis is an *inexact* science, because markets behave in ways that are very difficult to describe mathematically. However, probability analysis is also *necessary* for the option strategist; without it he would be in the dark as to the likelihood of profitable outcomes for his strategy. Overall, in a diversified set of positions, the option strategist should use the fat tail distribution in a Monte Carlo

simulation to estimate probabilities. However, if that is not available, he can use the normal or lognormal distribution with the proviso that he understands it is not “gospel.” He should require very stringent criteria on any strategies that are antivolatility strategies, such as naked option writing of stock options, for there is a greater than normal chance of a large move by the underlying, especially if the underlying is stock.

The sophisticated trader may want to view his probabilities in the light of more than one proposed distribution of prices. Of course, this type of analysis (using several distributions) puts the onus on the investor to choose the distribution that he wants to use in order to analyze his investment. However, such an approach should be extremely illustrative in that he can compare returns from different strategies and have a reasonable expectation as to which ones might perform the best under different market conditions.